

# Global Economics

## Grade 10 – Term 1 – Classroom activity 2

A decision under uncertainty selects an alternative  $A_i$  before a state  $S_j$  is known. Each pair  $(A_i, S_j)$  determines a payoff

$$P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j),$$

placed in row  $A_i$ , column  $S_j$ . The objective is to construct the complete payoff table.

Revenue and cost are

$$R = q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j),$$

$$C = q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j),$$

$$P_{ay}(A_i, S_j) = q(A_i, S_j)[r(A_i) + r(S_j) - c(A_i) - c(S_j)] + f_r(A_i) + f_r(S_j) - f_c(A_i) - f_c(S_j).$$

Variable components are per-unit rates multiplied by  $q(A_i, S_j)$ ; fixed components are not.  $A_i$ ,  $S_j$ , and  $(A_i, S_j)$  indicate dependence on the alternative, state, or both. Missing components are zero; directly given  $R$  or  $C$  require no decomposition.

### Context Descriptions

For each problem, **(C1)** *identify the decision alternatives, the states of nature, the uncertain event, and the consequence.* Then **(C3)** *calculate revenue, cost, and profit for every alternative–state pair and construct the payoff table.*

1. **Direct case.** A small café must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in thousand USD. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.
2. **One-component problem.** A custom-printing cooperative must choose between producing premium notebooks and producing canvas bags before market conditions are known. Conditions may be strong demand with stable material prices or weak demand with higher material prices. The respective quantities are 6 and 4 production lots for premium notebooks, and 5 and 3 lots for canvas bags. Revenue per lot depends on the product: 12 thousand USD for premium notebooks and 15 thousand USD for canvas bags. Cost per lot depends on the state: 5 thousand USD under strong demand and stable prices, and 8 thousand USD under weak demand and higher prices.
3. **Two-component problem.** A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 7 and 4 for the standard box and 6 and 3 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 8 and 11 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 6 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 9 thousand USD under strong demand and 13 thousand USD under weak demand.
4. **Three-component problem.** A regional logistics company must choose a standard or express delivery plan before demand and traffic conditions are known. Conditions may be high demand with light traffic or moderate demand with heavy traffic. Quantities, in hundreds of delivery batches, are 8 and 5 for the standard plan and 10 and 6 for the express plan under the two states, respectively. The plan-based revenue rates are 9 and 11 thousand USD per hundred batches; the state-based revenue rates are 3 and 2; and the plan-based fixed revenues are 16 and 24 thousand USD. The plan-based cost rates are 4 and 5 thousand USD per hundred batches; the state-based cost rates are 1 and 3; and the state-based fixed costs are 10 and 15 thousand USD.

# 1 Solutions: Revenue, Costs, Profit, and Payoff Tables

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**Problems:**

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- 1.1. Local Café Investment (C1–C3)

1.2. Custom Printing Product Line (C1–C3)

1.3. Farm Produce Subscription Plans (C1–C3)

1.4. Regional Parcel Delivery (C1–C3)
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## 1.1 Exercise 1: Local café investment (direct case)

### C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                            |                              |
|------------------|--------------------------------------------------------|------------------------------|
| Alternatives     | <b>A1:</b> Launch dessert line                         | <b>A2:</b> Keep current menu |
| States of nature | <b>S1:</b> High traffic                                | <b>S2:</b> Low traffic       |
| Events           | Actual foot traffic after the menu decision            |                              |
| Consequences     | Profit in thousand USD for each alternative–state pair |                              |



**C3 - Calculating profit and building the payoff table.** The total revenue and total cost are supplied directly for every cell, so

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

### Calculations (thousand USD).

| Alternative and state     | Revenue | Cost | Profit         |
|---------------------------|---------|------|----------------|
| Dessert line–high traffic | 58      | 16   | $58 - 16 = 42$ |
| Dessert line–low traffic  | 27      | 19   | $27 - 19 = 8$  |
| Current menu–high traffic | 49      | 21   | $49 - 21 = 28$ |
| Current menu–low traffic  | 41      | 21   | $41 - 21 = 20$ |

### Final payoff table (profit in thousand USD).

| Alternative         | High traffic | Low traffic |
|---------------------|--------------|-------------|
| Launch dessert line | 42           | 8           |
| Keep current menu   | 28           | 20          |

1.2 Exercise 2: Custom printing product line (one component)

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                                    |                                                   |
|------------------|----------------------------------------------------------------|---------------------------------------------------|
| Alternatives     | <b>A1:</b> Produce premium notebooks                           | <b>A2:</b> Produce canvas bags                    |
| States of nature | <b>S1:</b> Strong demand and stable material prices            | <b>S2:</b> Weak demand and higher material prices |
| Events           | Market conditions revealed after the product-line decision     |                                                   |
| Consequences     | Profit in thousand USD for each product-line–market-state pair |                                                   |



**C3 - Calculating profit and building the payoff table.** Revenue has one alternative-based variable component, while cost has one state-based variable component:

$$R = q(A_i, S_j)r(A_i), \quad C = q(A_i, S_j)c(S_j), \quad P = R - C.$$

**Calculations (thousand USD).**

| Cell         | Revenue      | Cost        | Profit         |
|--------------|--------------|-------------|----------------|
| $(A_1, S_1)$ | $6(12) = 72$ | $6(5) = 30$ | $72 - 30 = 42$ |
| $(A_1, S_2)$ | $4(12) = 48$ | $4(8) = 32$ | $48 - 32 = 16$ |
| $(A_2, S_1)$ | $5(15) = 75$ | $5(5) = 25$ | $75 - 25 = 50$ |
| $(A_2, S_2)$ | $3(15) = 45$ | $3(8) = 24$ | $45 - 24 = 21$ |

**Final payoff table (profit in thousand USD).**

| Alternative       | Strong demand, stable prices | Weak demand, higher prices |
|-------------------|------------------------------|----------------------------|
| Premium notebooks | 42                           | 16                         |
| Canvas bags       | 50                           | 21                         |

1.3 Exercise 3: Farm produce subscription plans (two components)

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                                              |                                        |
|------------------|--------------------------------------------------------------------------|----------------------------------------|
| Alternatives     | <b>A1:</b> Offer a standard produce box                                  | <b>A2:</b> Offer a premium produce box |
| States of nature | <b>S1:</b> Strong seasonal demand                                        | <b>S2:</b> Weak seasonal demand        |
| Events           | Seasonal market conditions revealed after the subscription-plan decision |                                        |
| Consequences     | Profit in thousand USD for each plan–market-state pair                   |                                        |



**C3 - Calculating profit and building the payoff table.** Revenue combines alternative-based and state-based variable components. Cost combines an alternative-based variable component and a state-based fixed component:

$$R = q(A_i, S_j)[r(A_i) + r(S_j)], \quad C = q(A_i, S_j)c(A_i) + f_c(S_j), \quad P = R - C.$$

Calculations (thousand USD).

| Cell         | Revenue          | Cost             | Profit         |
|--------------|------------------|------------------|----------------|
| $(A_1, S_1)$ | $7(8 + 3) = 77$  | $7(4) + 9 = 37$  | $77 - 37 = 40$ |
| $(A_1, S_2)$ | $4(8 + 1) = 36$  | $4(4) + 13 = 29$ | $36 - 29 = 7$  |
| $(A_2, S_1)$ | $6(11 + 3) = 84$ | $6(6) + 9 = 45$  | $84 - 45 = 39$ |
| $(A_2, S_2)$ | $3(11 + 1) = 36$ | $3(6) + 13 = 31$ | $36 - 31 = 5$  |

Final payoff table (profit in thousand USD).

| Alternative          | Strong seasonal demand | Weak seasonal demand |
|----------------------|------------------------|----------------------|
| Standard produce box | 40                     | 7                    |
| Premium produce box  | 39                     | 5                    |

1.4 Exercise 4: Regional parcel delivery (three components)

C1 - Interpreting decision alternatives, events, consequences, and states.

| Element          | Description                                                             |                                               |
|------------------|-------------------------------------------------------------------------|-----------------------------------------------|
| Alternatives     | <b>A1:</b> Use the standard delivery plan                               | <b>A2:</b> Use the express delivery plan      |
| States of nature | <b>S1:</b> High demand with light traffic                               | <b>S2:</b> Moderate demand with heavy traffic |
| Events           | Demand and traffic conditions revealed after the delivery-plan decision |                                               |
| Consequences     | Profit in thousand USD for each delivery-plan-state pair                |                                               |



**C3 - Calculating profit and building the payoff table.** Revenue has alternative- and state-based variable components and an alternative-based fixed component. Cost has alternative- and state-based variable components and a state-based fixed component:

$$\begin{aligned} R &= q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i), \\ C &= q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(S_j), \\ P &= R - C. \end{aligned}$$

Calculations (thousand USD).

| Cell         | Revenue                 | Cost                  | Profit          |
|--------------|-------------------------|-----------------------|-----------------|
| $(A_1, S_1)$ | $8(9 + 3) + 16 = 112$   | $8(4 + 1) + 10 = 50$  | $112 - 50 = 62$ |
| $(A_1, S_2)$ | $5(9 + 2) + 16 = 71$    | $5(4 + 3) + 15 = 50$  | $71 - 50 = 21$  |
| $(A_2, S_1)$ | $10(11 + 3) + 24 = 164$ | $10(5 + 1) + 10 = 70$ | $164 - 70 = 94$ |
| $(A_2, S_2)$ | $6(11 + 2) + 24 = 102$  | $6(5 + 3) + 15 = 63$  | $102 - 63 = 39$ |

Final payoff table (profit in thousand USD).

| Alternative            | High demand, light traffic | Moderate demand, heavy traffic |
|------------------------|----------------------------|--------------------------------|
| Standard delivery plan | 62                         | 21                             |
| Express delivery plan  | 94                         | 39                             |